

MM-Mixer: A Simple Yet Strong Baseline for Multimodal Emotion Recognition in Conversations

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Abstract

Multimodal Emotion Recognition in Conversation (MERC) aims to identify utterance-level emotions from the text, audio, and visual signals within a dialogue. Existing methods fuse unimodal representations and perform cross-modal interaction from a unified modality perspective, ignoring the utilization of complementary cross-modal interaction from different perspectives, such as the sequence dimension and the hidden feature dimension. Feature utilization from multiple perspectives can help the model capture sentiment cues from different modality signals more accurately. To address this issue, we propose a simple yet strong MERC framework named **MM-Mixer** to mine and model sentiment cues by performing cross-modal interaction and fusion from multiple perspectives. Specifically, the Adaptive Gating module is first deployed to extract modality-specific cues for subsequent modality-centric cross-modal alignment, which reduces irrelevant information before multimodal fusion. To extract multimodal sentiment cues from multiple perspectives, an Axis-wise Multimodal Mixer is proposed to perform cross-modal interaction along the latent sequence dimension, modality dimension, and hidden feature dimension in sequence. Finally, a simple fusion module enhanced with Explicit Pairwise Interaction Residual Correction is introduced to fuse all different modality features into a compact yet discriminative multimodal representation for emotion recognition. Extensive experiments on the IEMOCAP and MELD datasets show that our MM-Mixer consistently outperforms baselines in accuracy and weighted F1-score.

Introduction

Multimodal Emotion Recognition in Conversation (MERC) aims to identify the emotion expressed by each utterance in a dialogue by integrating textual, acoustic, and visual signals. Early ERC studies mainly focused on text-based conversational context and speaker dependencies through recurrent state tracking and graph-based reasoning (Majumder et al. 2019; Ghosal et al. 2019). With the increasing availability of multimodal conversational data, subsequent MERC methods have incorporated acoustic and visual signals through multimodal graph learning (Hu et al. 2021), cross-modal

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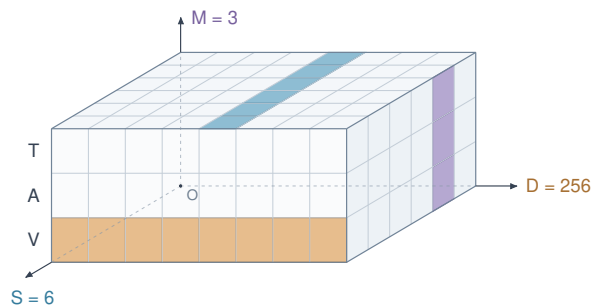


Figure 1: Tensor organization in AMM. Three aligned modality branches are expanded into S learned projection views with D hidden features. Colored cell groups highlight example vectors along the projection (S), modality (M), and hidden-feature (D) axes, with the other two indices fixed. The feature axis is drawn schematically. Learned views are not temporal steps.

attention (Mao et al. 2021), and more expressive fusion mechanisms (Ma et al. 2024; Lyu et al. 2026). These approaches have substantially advanced multimodal representation learning for MERC by enabling information exchange among heterogeneous modality streams.

Figure 1 previews the axis-wise interaction used in our proposed framework.

Despite this progress, fusion designs that treat each utterance-level modality-specific representation as a holistic interaction unit leave the organization of multiple learned views within each modality implicit. Although such representations may contain rich affective information, exchanging and aggregating these vectors does not by itself distinguish within-modality view processing from cross-modal exchange. A remaining design question is how to construct multiple learned views and organize these two types of interaction along distinct representation axes.

To address this limitation, we propose MM-Mixer, a simple yet strong framework that organizes multimodal interaction through multiple learned projection views. MM-Mixer first applies an Adaptive Gating module that uses a shared multimodal reference to calibrate modality-specific features,

followed by modality-centric cross-modal alignment to prepare the branch representations for subsequent multi-view interaction.

Inspired by the multiple-projection design of multi-head attention (Vaswani et al. 2017), the core Axis-wise Multimodal Mixer transforms each aligned branch representation into a fixed-length set of learned projection views. These views form the latent sequence used in the mixer and do not correspond to temporally ordered conversational utterances. The projection-axis mixing captures dependencies among views within each modality branch. The modality-axis mixing exchanges information among textual, acoustic, and visual branches, while the hidden-feature-axis mixing refines the channels of each view. This structure assigns within-branch view mixing, cross-modal exchange, and hidden-feature transformation to dedicated operations.

To obtain a compact yet discriminative multimodal representation for emotion recognition, a query-based fusion module aggregates the mixer outputs into the main fused representation. Explicit Pairwise Interaction Residual Correction complements this pathway with low-rank pairwise multiplicative interactions, which are projected and added to the main fused representation as a residual correction.

Extensive experiments on popular MERC benchmarks, IEMOCAP (Busso et al. 2008) and MELD (Poria et al. 2019), under a unified history-only protocol, where future conversational information is excluded during prediction, show that MM-Mixer achieves impressive performance in accuracy and weighted F1-score against representative baselines. Ablation studies further examine the effects of the Adaptive Gating module, Axis-wise Multimodal Mixer module, and fusion module enhanced with Explicit Pairwise Interaction Residual Correction.

Our main contributions are summarized as follows:

- We propose MM-Mixer, a simple yet strong MERC framework that models multimodal emotion cues through progressive modality refinement and multi-dimensional interaction.
- We introduce an Axis-wise Multimodal Mixer to extract multimodal sentiment cues from multiple perspectives by performing multi-dimensional feature interaction along the latent-sequence, modality, and hidden-feature dimensions.
- We develop an Explicit Pairwise Interaction Residual Correction that captures pairwise dependencies across modalities and incorporates them as residual corrections to the main fused representation.
- Extensive experiments on popular MERC benchmarks, IEMOCAP and MELD, under a unified history-only protocol show that MM-Mixer achieves impressive performance in accuracy and weighted F1-score against representative baselines.

Related Work

Context Modeling in ERC. Conversational context provides important evidence for interpreting an utterance whose emotion cannot be determined in isolation. Sequence-based methods model this evidence by tracking dialogue and

speaker states. For example, DialogueRNN (Majumder et al. 2019) maintains speaker-specific states together with a global dialogue state. Graph-based methods instead represent utterances as nodes and propagate information through speaker-dependent relations, as in DialogueGCN (Ghosal et al. 2019). DAG-ERC (Shen et al. 2021) further organizes information flow with a directed acyclic graph that combines local and long-range context. These methods mainly differ in how information is propagated across utterances and speakers. More recently, PRC-Emo (Li et al. 2026) combines emotion-sensitive prompting, demonstration retrieval, and a curriculum based on weighted emotion shifts between same- and different-speaker utterances to model explicit and implicit contextual cues.

MM-Mixer does not introduce a new dialogue topology and focuses on multimodal interaction after the causal textual context has been encoded.

Multimodal Interaction and Fusion. MERC methods must combine heterogeneous textual, acoustic, and visual signals while retaining information specific to each modality. MMGCN (Hu et al. 2021) constructs a multimodal graph to model cross-modal and speaker-dependent relations, while MMDAG (Xu et al. 2022) extends directed acyclic context modeling to multimodal inputs. Transformer-based approaches use intra- and inter-modal attention for direct information exchange. DialogueTRM (Mao et al. 2021) follows this strategy to model multimodal emotional dynamics, and SDT (Ma et al. 2024) combines intra- and inter-modal Transformers with gated fusion and unimodal self-distillation. Other methods introduce task-specific interaction structures. ECERC (Zhang and Tan 2025) separates target-utterance evidence from contextual causes, whereas GS-MCC (Ai et al. 2025) models multimodal conversations through graph-spectral operators. Recent work has also moved toward asymmetric or component-aware fusion. EMART (Seong et al. 2026) treats text as the primary information source and audio as a complementary signal, coupling modality-aware fusion with an emotion-wheel-guided supervised contrastive objective. DnR (Mai, Nguyen, and Le 2026) explicitly decomposes multimodal representations into unique, pairwise redundant, and synergistic components before applying component-specific refinement. Together, these methods organize multimodal reasoning through graph propagation, attention-based exchange, or task-specific interaction structures.

Compared with these methods, MM-Mixer organizes multimodal interaction according to the internal structure of the utterance-level representation. It assigns dedicated transformations to the learned-sequence, modality, and hidden-feature axes, enabling structured information exchange with a lightweight mixer architecture.

Axis-Factorized and Multiplicative Interaction. Axis-factorized networks assign different transformations to different dimensions of a representation. MLP-Mixer (Tolstikhin et al. 2021) separates token mixing from channel mixing, demonstrating that interactions along these two axes can be modeled by dedicated MLPs. Multiplicative fusion provides another way to expose cross-modal dependencies. Factorized bilinear pooling and low-rank multimodal fusion approxi-

mate tensor interactions without constructing a full outer-product tensor (Yu et al. 2017; Liu et al. 2018). In MERC, CSS (Lyu et al. 2026) uses low-rank polynomial fusion as its main multimodal fusion operator and coordinates multiple objectives through Pareto gradient modulation.

Methodology

MM-Mixer Overview

Given a dialogue $\{u_t\}_{t=1}^N$ containing N utterances, MERC aims to predict an emotion label $y_t \in \{1, \dots, C\}$ for each utterance u_t . The model input contains a textual feature $\mathbf{x}_t^T \in \mathbb{R}^{d_T}$, an acoustic feature $\mathbf{x}_t^A \in \mathbb{R}^{d_A}$, and a visual feature $\mathbf{x}_t^V \in \mathbb{R}^{d_V}$. The superscripts T , A , and V identify the textual, acoustic, and visual modalities, respectively. The textual feature represents the target utterance together with its preceding dialogue context, whereas the acoustic and visual features describe the target utterance. All inputs exclude utterances after u_t . MM-Mixer learns a mapping f_θ from the three modality features to the emotion space and selects the class with the highest predicted probability:

$$\hat{y}_t = \underset{c \in \{1, \dots, C\}}{\operatorname{argmax}} f_\theta(\mathbf{x}_t^T, \mathbf{x}_t^A, \mathbf{x}_t^V)_c \quad (1)$$

where f_θ denotes the model parameterized by θ , and $\hat{y}_t$ is the predicted emotion label of u_t . For clarity, the utterance index t is omitted below.

As illustrated in Figure 2, MM-Mixer first employs pre-trained modality-specific encoders to extract utterance-level unimodal features and projects them into a common feature space. Modality-specific feature gates suppress irrelevant channels. Adaptive Gating (AG) then constructs a shared multimodal reference and uses it to calibrate each modality, after which Modality-centric Cross-modal Alignment (MCA) uses each modality to query complementary information from the other two modalities. The aligned representations are then organized along the learned-sequence, modality, and hidden-feature axes by the Axis-wise Multimodal Mixer (AMM). Finally, query-based integration forms the main fusion pathway, while the Explicit Pairwise Interaction Residual Correction (EPIRC) branch provides an additive low-rank multiplicative correction. The fused prediction and three unimodal predictions are jointly optimized during training.

Unimodal Feature Extraction

MM-Mixer first deploys pre-trained modality-specific encoders to extract utterance-level unimodal features $\mathbf{x}^T$, $\mathbf{x}^A$, and $\mathbf{x}^V$. Since the input modalities have different feature dimensions and feature distributions, each modality is projected into a common feature space to reduce their heterogeneity, followed by filtering irrelevant information with a feature gating as follows:

$$\begin{aligned} \mathbf{e}^m &= \text{Dropout}(\text{GELU}(\text{LN}(\mathbf{W}_p^m \mathbf{x}^m + \mathbf{b}_p^m))) \\ \text{coff}^m &= \sigma(\mathbf{W}_{g2}^m \text{ReLU}(\text{LN}(\mathbf{W}_{g1}^m \mathbf{e}^m + \mathbf{b}_{g1}^m)) + \mathbf{b}_{g2}^m) \\ \mathbf{u}^m &= \mathbf{e}^m \odot \text{coff}^m \end{aligned} \quad (2)$$

where $\mathbf{e}^m \in \mathbb{R}^D$ is the unimodal representation, $\mathbf{u}^m \in \mathbb{R}^D$ is its refined version, and $m \in M$, with $M = \{T, A, V\}$. $\mathbf{W}_p^m$, $\mathbf{W}_{g1}^m$ and $\mathbf{W}_{g2}^m$ are learnable weights. $\mathbf{b}_p^m$, $\mathbf{b}_{g1}^m$ and $\mathbf{b}_{g2}^m$ are learnable biases. Dropout, GELU, ReLU, and LN denote the dropout, GELU activation, RELU activation, and layer normalization, respectively. σ denotes the Sigmoid function. Here, it is used to suppress irrelevant channels while preserving one feature vector per modality.

Adaptive Gating

Given the filtered unimodal features, the AG module constructs a shared multimodal reference as follows:

$$\begin{aligned} \alpha &= \text{softmax}(\text{MLP}([\mathbf{u}^T; \mathbf{u}^A; \mathbf{u}^V])) \\ \mathbf{c} &= \sum_{m \in M} \alpha_m \mathbf{u}^m \end{aligned} \quad (3)$$

where $\alpha \in \mathbb{R}^3$ contains the normalized modality coefficients, and $\mathbf{c} \in \mathbb{R}^D$ is their weighted multimodal reference. The shared reference then conditions a separate channel gate to calibrate modality features for each modality:

$$\mathbf{f}^m = [\sigma(\text{MLP}_c^m([\mathbf{u}^m; \mathbf{c}])) + \epsilon_g] \odot \mathbf{u}^m \quad (4)$$

Here, $\mathbf{f}^m \in \mathbb{R}^D$ is the calibrated modality feature. The context-conditioned sigmoid coefficients perform channel selection, while ϵ_g preserves a direct contribution from $\mathbf{u}^m$.

Modality-centric Cross-modal Alignment

The MCA module comprises three directional cross-attention branches. For each $m \in M$, one branch uses the calibrated feature of modality m as its query, while the other two modalities provide the keys and values:

$$\begin{aligned} \mathbf{K}^m &= \mathbf{V}^m = \text{Stack}\{\mathbf{f}^n \mid n \in M, n \neq m\} \\ \mathbf{h}^m &= \text{MHA}^m(\mathbf{f}^m, \mathbf{K}^m, \mathbf{V}^m) + \epsilon_a \mathbf{f}^m \end{aligned} \quad (5)$$

Here, $\mathbf{K}^m$ and $\mathbf{V}^m \in \mathbb{R}^{2 \times D}$ contain the two peer-modality vectors. MHA^m is the multi-head attention layer for query modality m , and ϵ_a controls the query residual. The output $\mathbf{h}^m \in \mathbb{R}^D$ is the cross-modal aligned feature of modality m .

Axis-wise Multimodal Mixer

AMM aims to mine the sentiment cues along different dimensions from the cross-modal aligned feature $\mathbf{h}^m$, since the three axes of a multimodal representation have different meanings: learned elements within a modality, modality identity, and hidden feature channels. AMM processes these axes separately so that each operation has a defined input dimension and interaction scope.

Figure 3 illustrates the learned projection views and their interaction scopes. The projection axis indexes different learned transformations of each aligned branch representation, rather than observed temporal frames or predefined semantic roles.

Sequence construction. For every modality, a shared linear map expands one D -dimensional vector $\mathbf{h}^m$ into a sequence of S learned elements, and then these sequences are stacked as a complete multimodal tensor $\mathbf{Z}^0 \in \mathbb{R}^{3 \times S \times D}$:

$$\mathbf{Z}^0 = \text{Stack}_{m \in M} [\text{reshape}_{S \times D}(\mathbf{W}_s \mathbf{h}^m + \mathbf{b}_s)] \quad (6)$$

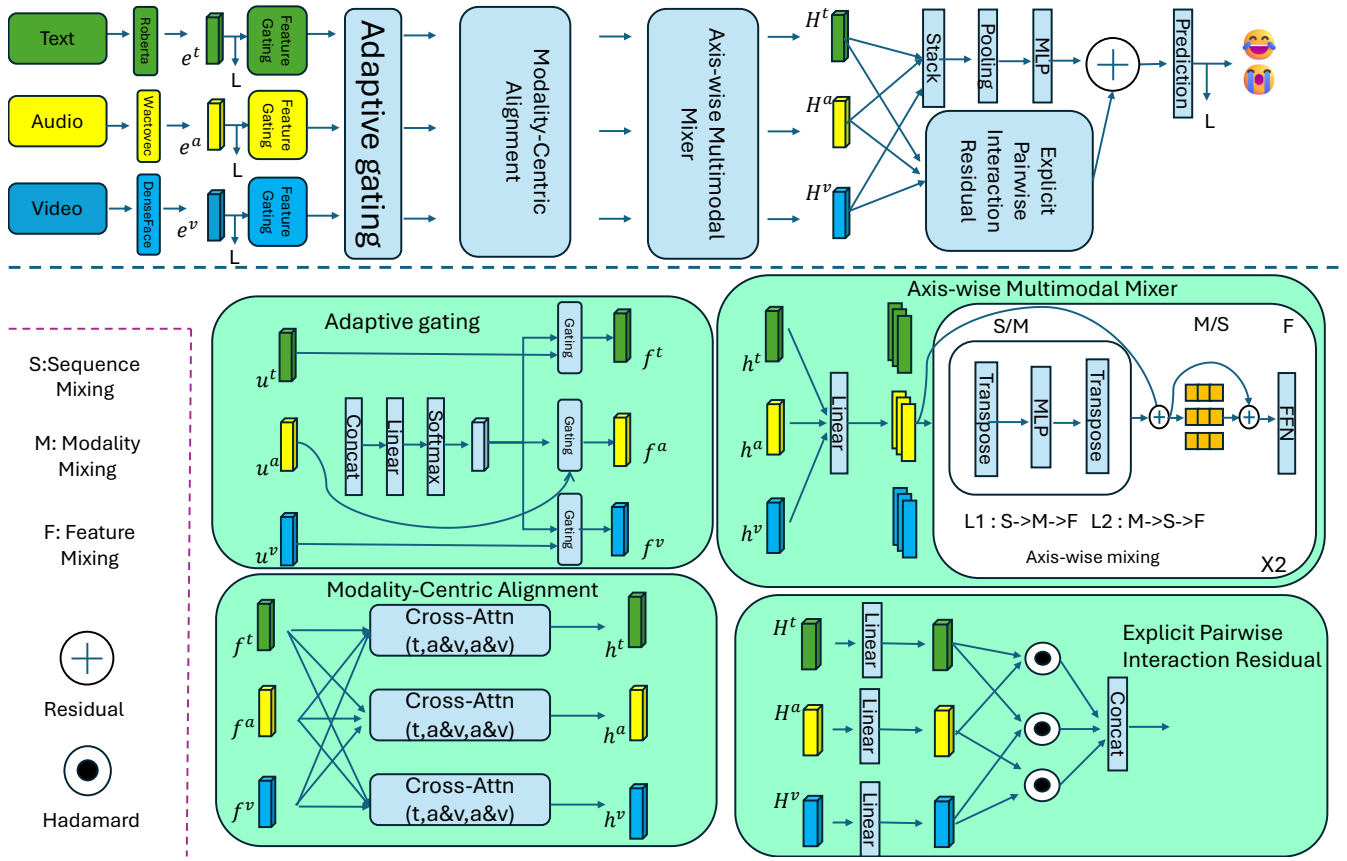


Figure 2: Overview of MM-Mixer. Pre-extracted textual, acoustic, and visual features are projected and gated before adaptive refinement and modality-centric cross-modal alignment. Two alternating axis-wise mixer blocks organize the resulting representations along the latent-sequence, modality, and feature dimensions. Query-based integration forms the main prediction pathway, while explicit low-rank pairwise interactions provide a residual correction. Layer normalization and internal residual connections are omitted for clarity.

where the parameters $\mathbf{W}_s$ and $\mathbf{b}_s$ are shared by T , A , and V , giving corresponding sequence coordinates a common parameterization. The modality axis of $\mathbf{Z}^0$ follows the fixed order V, A, T .

For the sake of consistency, the notation $[\cdot]_{m,s,d}$ denotes an entry at modality m , learned-sequence element s , and feature channel d in the remaining part.

Axis-wise mixing. Each mixer block contains three pre-normalized residual operators. Let $\mathbf{Z} \in \mathbb{R}^{3 \times S \times D}$ denote a generic input to an axis-mixing operator. Sequence mixing applies an MLP along the S axis while holding the modality and feature indices fixed:

$$[\mathcal{S}(\mathbf{Z})]_{m,:,d} = [\mathbf{Z}]_{m,:,d} + \text{MLP}_S([\text{LN}_S(\mathbf{Z})]_{m,:,d}). \quad (7)$$

Here, MLP_S maps the sequence axis through a hidden layer and returns an output of the same width.

Modality mixing instead holds (s, d) fixed and transforms the three-vector $[[\mathbf{Z}]_{V,s,d}; [\mathbf{Z}]_{A,s,d}; [\mathbf{Z}]_{T,s,d}]$:

$$[\mathcal{M}(\mathbf{Z})]_{:,s,d} = [\mathbf{Z}]_{:,s,d} + \mathbf{W}_r[\text{LN}_M(\mathbf{Z})]_{:,s,d}. \quad (8)$$

The bias-free routing matrix $\mathbf{W}_r \in \mathbb{R}^{3 \times 3}$ is shared across all sequence coordinates and feature channels, providing a com-

mon cross-modal transformation at every sequence-feature coordinate.

Feature mixing completes the block by applying a channel-wise MLP to every modality-sequence pair:

$$[\mathcal{F}(\mathbf{Z})]_{m,s,:} = [\mathbf{Z}]_{m,s,:} + \text{MLP}_D([\text{LN}_F(\mathbf{Z})]_{m,s,:}). \quad (9)$$

Here, $\mathcal{S}$, $\mathcal{M}$, and $\mathcal{F}$ denote sequence, modality, and feature mixing, respectively. The feature MLP follows $D \rightarrow D_h \rightarrow D$ with GELU, where D_h is its hidden width.

We stack two mixer blocks with different operation orders:

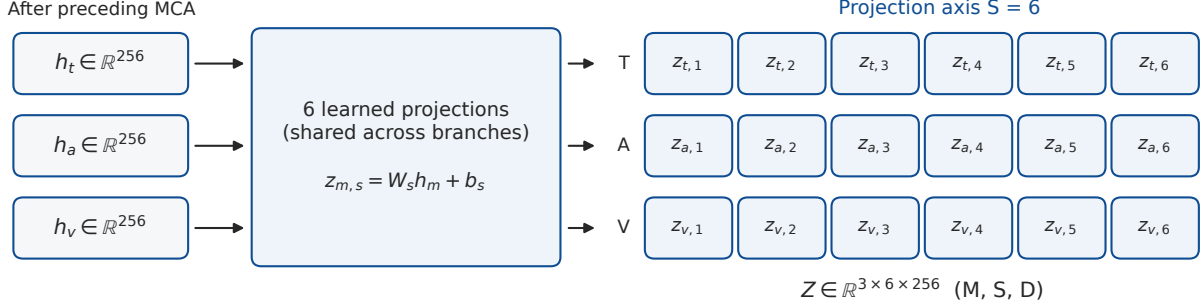
$$\begin{aligned} \mathbf{Z}^1 &= (\mathcal{F}^1 \circ \mathcal{M}^1 \circ \mathcal{S}^1)(\mathbf{Z}^0) \\ \mathbf{Z}^2 &= (\mathcal{F}^2 \circ \mathcal{S}^2 \circ \mathcal{M}^2)(\mathbf{Z}^1). \end{aligned} \quad (10)$$

$\mathbf{Z}^\ell$ denotes the output tensor of mixer block ℓ , and the composition symbol $\circ$ is evaluated from right to left. Therefore, the first mixer block organizes sequence elements before routing across modalities, whereas the second routes modalities before sequence refinement. Averaging over the sequence axis returns one vector per modality:

$$\mathbf{H}^m = \frac{1}{S} \sum_{s=1}^S \mathbf{Z}_{m,s,:}^2 \quad (11)$$

AMM: learned projection views and axis-wise interaction

a Expand each branch representation



b Mix axes, then aggregate

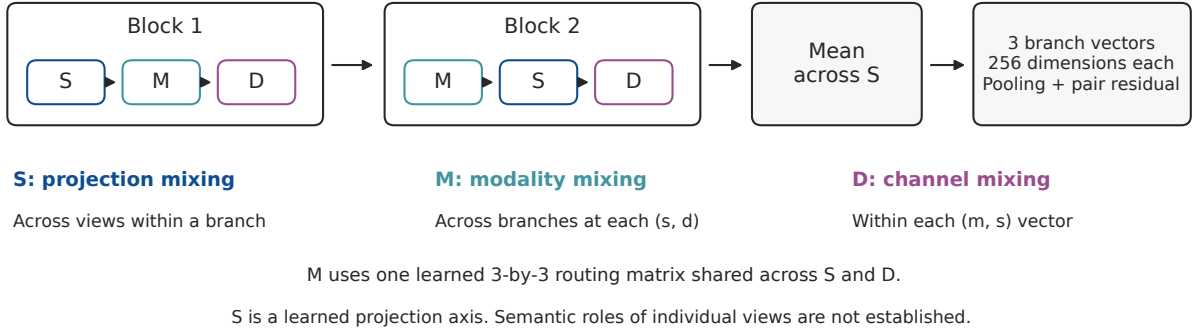


Figure 3: Learned projection views and axis-wise interaction in AMM. Each 256-dimensional branch representation after MCA is expanded into six learned projections shared across branches. Two blocks use projection–modality–channel and modality–projection–channel update orders, followed by averaging over the projection axis. Modality mixing uses one learned 3×3 routing matrix shared across projection/channel positions. Layer normalization and residual additions are omitted for clarity; individual projections have no predefined semantic roles.

$\mathbf{H}^m \in \mathbb{R}^D$ is the mixer output for modality m .

Multimodal Fusion with Explicit Pairwise Interaction Residual Correction

Given the three modality representations produced by AMM, the multimodal fusion contains a main fusion path and a parallel residual path. The main path first stacks the mixer outputs along the modality axis:

$$\mathbf{H} = \text{Stack}(\mathbf{H}^V, \mathbf{H}^A, \mathbf{H}^T) \quad (12)$$

where $\mathbf{H} \in \mathbb{R}^{3 \times D}$. Three learnable queries then aggregate the stacked modality representations via multi-head attention:

$$\mathbf{P} = \text{MHA}(\mathbf{Q}_p, \mathbf{H}, \mathbf{H}) \quad (13)$$

Here, $\mathbf{Q}_p \in \mathbb{R}^{3 \times D}$ contains three learned query vectors, and $\mathbf{P}$ contains their pooled outputs. The outputs are flattened and integrated to form the main fused representation $\mathbf{h}_{\text{main}} \in \mathbb{R}^D$:

$$\mathbf{h}_{\text{main}} = \text{MLP}_{\text{int}}(\text{vec}(\mathbf{P})) \quad (14)$$

where $\text{vec}(\cdot)$ denotes the vectorization operation. The integration MLP maps the flattened $3D$ -dimensional representation back to the common feature space, where $\text{vec}(\cdot)$ denotes the vectorization operation.

Although the main fusion path adaptively aggregates the three modality representations, it models their relations implicitly. To make the fused representation more discriminative, we introduce EPIRC as a parallel residual path that explicitly captures multiplicative relations between modality pairs and uses them to correct the main fused representation. Each mixer output is first projected into a rank- R space:

$$\mathbf{p}^m = \mathbf{W}_x^m \mathbf{H}^m + \mathbf{b}_x^m \quad (15)$$

where $\mathbf{p}^m \in \mathbb{R}^R$ and R is the interaction rank and $\mathbf{p}^m$ is the low-rank representation of modality m . Then, the explicit pairwise interaction between two modalities is modeled as

the product between two mixer outputs as follows:

$$\begin{aligned} \mathbf{r}_{AV} &= \mathbf{p}^A \odot \mathbf{p}^V \\ \mathbf{r}_{TV} &= \mathbf{p}^T \odot \mathbf{p}^V \\ \mathbf{r}_{TA} &= \mathbf{p}^T \odot \mathbf{p}^A \end{aligned} \quad (16)$$

where $\mathbf{r}_{AV}$, $\mathbf{r}_{TV}$, and $\mathbf{r}_{TA}$ encode the three modality-pair dependencies. After obtaining the interactions $\mathbf{r}_{AV}$, $\mathbf{r}_{TV}$, and $\mathbf{r}_{TA}$, the residual correction $\mathbf{r} \in \mathbb{R}^D$ can be computed as follows:

$$\begin{aligned} \mathbf{r}_{\text{cat}} &= [\mathbf{r}_{AV}; \mathbf{r}_{TV}; \mathbf{r}_{TA}] \\ \mathbf{r} &= \mathbf{W}_o \mathbf{r}_{\text{cat}} + \mathbf{b}_o \in \mathbb{R}^D \end{aligned} \quad (17)$$

where $;$ denotes the concatenation operation. $\mathbf{W}_o$ and $\mathbf{b}_o$ are initialized to zero, so this residual correction branch starts with a zero output and progressively learns a pairwise correction without perturbing the initial main fusion path.

With $\mathbf{h}_{\text{main}}$ and $\mathbf{r}$, the final fused multimodal representation $\mathbf{h} \in \mathbb{R}^D$ can be obtained as follows:

$$\mathbf{h} = \mathbf{h}_{\text{main}} + \mathbf{r} \quad (18)$$

Emotion Prediction and Learning Objective

To predict emotion for the current utterance, a linear classifier is used to map the final fused multimodal representation $\mathbf{h}$ to the logit distribution $\mathbf{z} \in \mathbb{R}^C$ of the emotion label space:

$$\mathbf{z} = \mathbf{W}_c \mathbf{h} + \mathbf{b}_c \quad (19)$$

where $\mathbf{W}_c$ and $\mathbf{b}_c$ are the learnable weights and bias of the linear classifier. To optimize our model, we deploy Poly-Loss (Leng et al. 2022):

$$\mathcal{L}_{\text{main}} = \frac{1}{B} \sum_{i=1}^B \omega_i \ell_{\text{poly}}(\mathbf{z}_i, y_i; \mathbf{w}) \quad (20)$$

Here, ω_i denotes the focal modulation coefficient and B is the number of samples. $\mathbf{z}_i$ is the fused multimodal emotion logits of sample i in Eq. (19). ℓ_{poly} is per-sample Poly loss, which can be defined as follows:

$$\ell_{\text{poly}}(\mathbf{z}, y; w_y) = -w_y \log p_y + \alpha_p (1 - p_y)^{\gamma_p + 1} \quad (21)$$

where $p_y = \text{softmax}(\mathbf{z})_y$ is the probability of target class y , w_y is the class weight. α_p and γ_p control the Poly term.

Beyond the $\mathcal{L}_{\text{main}}$, we also deploy an auxiliary loss $\mathcal{L}_{\text{aux}}$ to help model learning. We add three auxiliary classifiers directly on the post-feature-gating vectors $\mathbf{u}^T$, $\mathbf{u}^A$, and $\mathbf{u}^V$, before adaptive context construction. For sample i and modality m , the corresponding auxiliary logits are

$$\mathbf{z}_{\text{aux},i}^m = \text{MLP}_{\text{aux}}^m(\mathbf{u}_i^m) \quad (22)$$

where $\text{MLP}_{\text{aux}}^m$ is a modality-specific two-layer classifier with GELU and dropout, and $\mathbf{z}_{\text{aux},i}^m \in \mathbb{R}^C$ contains its unimodal emotion logits. These classifiers are used only during training to supervise the output of feature gating, encouraging the gated representations to retain modality-specific discriminative information before cross-modal interaction.

Thus, the auxiliary loss $\mathcal{L}_{\text{aux}}$ can be constructed by aggregating the unimodal Poly loss of each modality:

$$\mathcal{L}_{\text{aux}} = \sum_{m \in \{T, A, V\}} \lambda_m \frac{1}{B} \sum_{i=1}^B \ell_{\text{poly}}(\mathbf{z}_{\text{aux},i}^m, y_i; \mathbf{w}_m) \quad (23)$$

Therefore, the total learning objective is defined as the weighted sum of $\mathcal{L}_{\text{main}}$ and $\mathcal{L}_{\text{aux}}$:

$$\mathcal{L} = \lambda_{\text{main}} \mathcal{L}_{\text{main}} + \mathcal{L}_{\text{aux}} \quad (24)$$

Experiments

Experiment Settings

Datasets. We evaluate MM-Mixer on two widely used MERC benchmarks: IEMOCAP (Busso et al. 2008) and MELD (Poria et al. 2019). IEMOCAP contains approximately 12 hours of dyadic conversations with synchronized transcriptions, speech, and visual behavior. There are six classes of data in IEMOCAP, including happiness, sadness, neutral, anger, excitement, and frustration. MELD is derived from the Friends TV series and contains multi-party conversations annotated at the utterance level with seven emotion categories. It has more than 1400 dialogues and 13000 utterances with seven emotion annotations: anger, disgust, sadness, joy, neutral, surprise, and fear. We follow the standard train, validation, and test splits and classification settings for both datasets.

Baselines. We compare MM-Mixer with representative methods from three mainstream MERC paradigms. First, DialogueRNN (Majumder et al. 2019) is a sequence-based model that tracks dialogue context and speaker states with recurrent networks. Second, MMGCN (Hu et al. 2021), MM-DFN (Hu et al. 2022), and M3Net (Chen et al. 2023) are graph-based methods that model structural dependencies among utterances and modalities. Third, Ada2I (Nguyen et al. 2024), SDT (Ma et al. 2024), and CSS (Lyu et al. 2026) employ attention-based or specialized fusion mechanisms to exchange and integrate multimodal information. Together, these baselines cover recurrent context modeling, graph reasoning, and multimodal attention and fusion. For fair comparison, we retain their models with their released data, features, and hyperparameters to avoid introducing method-specific changes beyond the evaluated protocol.

Evaluation metrics. We report per-class F1, accuracy (ACC), and weighted F1 (WF1). ACC and WF1 serve as the main overall metrics. Evaluation is conducted under a history-only evaluation protocol in which, when predicting the emotion of utterance u_t , a model can only take u_t and the dialogue history $\{u_1, \dots, u_{t-1}\}$ as input. We use the full dialogue history, up to 109 preceding utterances on IEMOCAP and 32 on MELD.

Implementation Details. For each utterance, we use RoBERTa-large (Liu et al. 2019), wav2vec 2.0 (Baevski et al. 2020), and DenseFace (Zhao et al. 2018) to obtain the textual, acoustic, and visual representations, respectively. Their dimensions are 1024, 1024, and 342. All modality features are projected to a common width of $D = 256$. The Modality-centric Cross-modal Alignment and learnable-query pooling

Model	Hap.	Sad.	Neu.	Ang.	Exc.	Fru.	ACC	WF1
DialogueRNN	37.25±4.53	70.94±0.80	53.94±1.12	60.40±0.95	64.05±1.44	59.05±0.47	58.68±0.23	58.77±0.12
MMGCN	36.07±3.02	68.65±0.39	50.63±1.79	62.65±0.55	65.48±3.88	60.70±1.11	59.31±0.96	58.86±0.87
MM-DFN	40.61±1.12	75.33±0.30	59.06±2.71	64.92±0.85	70.51±1.87	62.81±1.33	63.60±0.76	63.53±0.68
M3Net	41.71±1.79	72.92±3.11	62.33±0.91	64.53±1.06	75.62±0.93	57.72±2.25	63.96±0.37	63.70±0.30
Ada2I	48.90±5.40	79.62±0.85	60.37±3.04	61.53±3.29	72.26±1.47	64.42±1.24	65.70±1.63	65.52±1.70
SDT	62.27 ±1.41	78.32±0.77	71.46±0.48	67.90±1.01	79.21 ±0.96	65.04±1.36	71.19±0.38	71.23±0.48
CSS	<u>61.58</u> ±2.51	<u>79.74</u> ±0.67	72.27 ±0.48	<u>68.70</u> ±0.48	<u>77.22</u> ±0.31	<u>66.00</u> ±0.56	<u>71.51</u> ±0.15	<u>71.51</u> ±0.23
MM-Mixer	57.72±0.22	84.14 ±0.22	<u>71.81</u> ±0.36	71.59 ±1.06	74.04±1.12	69.15 ±0.38	72.07 ±0.15	72.19 ±0.12

Table 1: Performance comparison on the IEMOCAP dataset. All entries are reported as mean $\pm$ standard deviation in percentages. The best and second-best means are shown in bold and underlined, respectively.

Model	Neu.	Sur.	Fea.	Sad.	Joy	Dis.	Ang.	ACC	WF1
DialogueRNN	76.00±0.81	48.49±0.88	0.00±0.00	24.02±2.41	51.51±0.70	0.00±0.00	45.12±1.26	59.63±0.75	57.60±0.21
MMGCN	76.52±0.27	44.64±1.76	0.00±0.00	17.32±6.40	53.53±0.90	0.00±0.00	44.44±0.69	60.56±1.19	57.47±0.16
MM-DFN	74.75±2.19	47.76±2.66	0.94±1.33	17.88±5.49	51.73±1.72	0.00±0.00	42.18±2.43	60.29±0.39	57.95±0.24
M3Net	79.43±0.11	57.70±0.26	<u>21.79</u> ±3.65	37.45±1.39	63.89±0.87	23.67±4.27	53.68±0.96	66.78±0.17	65.39±0.22
Ada2I	77.29±0.40	49.46±1.45	0.00±0.00	25.56±3.07	56.19±0.84	0.00±0.00	48.82±0.56	62.59±0.59	59.66±0.29
SDT	79.85±0.16	<u>58.13</u> ±0.25	17.20±1.18	<u>42.64</u> ±1.92	63.95±0.20	28.34±2.01	53.63±0.59	67.06±0.44	66.09±0.39
CSS	<u>80.04</u> ±0.02	57.90±0.21	19.28±1.97	41.71±0.77	<u>64.10</u> ±0.34	<u>29.97</u> ±0.74	<u>54.36</u> ±0.46	<u>67.25</u> ±0.11	<u>66.28</u> ±0.13
MM-Mixer	80.70 ±0.16	59.79 ±0.45	29.57 ±1.94	42.87 ±0.59	65.34 ±0.59	34.93 ±3.86	55.94 ±0.55	68.36 ±0.25	67.62 ±0.23

Table 2: Performance comparison on the MELD dataset. All entries are reported as mean $\pm$ standard deviation in percentages. The best and second-best means are shown in bold and underlined, respectively.

Model	IEMOCAP	MELD
DialogueRNN	8.88	2.67
MMGCN	2.70	2.55
MM-DFN	2.06	2.87
M3Net	6.55	14.11
Ada2I	35.67	18.56
SDT	79.69	78.39
CSS	50.45	53.40
MM-Mixer	6.36	5.71

Table 3: Registered downstream parameters (millions), excluding pretrained feature extractors. Model configurations correspond to those profiled in Table 4.

Model	IEMOCAP	MELD
DialogueRNN	42.78	53.97
MMGCN	482.41	54.93
MM-DFN	351.20	131.49
M3Net	11.71	29.01
Ada2I	124.89	87.94
SDT	289.80	385.88
CSS	200.96	286.10
MM-Mixer	69.00	68.72

Table 4: Downstream computational cost on IEMOCAP and MELD, measured in MFLOPs per utterance.

use eight attention heads. The Axis-wise Multimodal Mixer constructs $S = 6$ learned elements for each modality. Its hidden-feature MLP has a hidden width of 1536. The Explicit Pairwise Interaction Residual Correction uses an interaction rank of $R = 32$. We optimize MM-Mixer with AdamW using base learning rates of 3×10^{-5} on IEMOCAP and 4.17×10^{-5} on MELD. The weight decay is 2×10^{-4} and 9.98×10^{-5} , respectively. We use a batch size of 32 and train for up to 100 epochs on IEMOCAP and 50 epochs on MELD. The dropout rate is set to 0.2 on both datasets. The Poly-loss parameters are $\alpha_p = 1.951$ and $\gamma_p = 1.402$, with focal exponent $\gamma_f = 2.5$. On IEMOCAP, the coefficients ($\lambda_{\text{main}}, \lambda_T, \lambda_A, \lambda_V$) are (0.45, 0.33, 0.11, 0.11). On MELD, the fused and three unimodal objectives are equally weighted. All experiments are run on an NVIDIA TESLA V100 with three seeds [0, 1, 42], and we report their average performance and standard deviation.

Main Results

Performance comparison. The experimental results on the IEMOCAP benchmark are shown in Table 1. We can see that MM-Mixer achieves the highest ACC (72.07%) and WF1 (72.19%). It outperforms the CSS model by 0.56% and 0.68% in ACC and WF1, respectively. Specifically, compared with CSS, it improves sadness by 4.40%, anger by 2.89%, and frustration by 3.15%. These results suggest that the axis-wise interaction and explicit pairwise interaction residual correction are effective for recognizing emotions whose evidence is distributed across multiple modalities. MM-Mixer cannot dominate every category, but it obtains the best F1 scores for sadness, anger, and frustration and provides competitive performance on happiness, neutral, and excitement. This pattern indicates that the overall improvement does not arise from uniformly favoring a single emotion category.

The experimental results on the MELD benchmark are shown in Table 2. The seven-class MELD benchmark is larger and more imbalanced, but MM-Mixer achieves the

Model Variant	IEMOCAP		MELD	
	WF1	ACC	WF1	ACC
Full MM-Mixer	72.16	72.03	68.01	68.77
w/o AG	71.68	71.53	67.36	68.08
w/o MCA	71.13	70.98	67.67	68.35
w/o AMM	71.44	71.29	67.05	67.60
w/o EPIRC	71.64	71.47	67.10	67.82
w/o Feature Gating	71.90	71.78	67.67	68.44
w/o Sequence Mixing	71.64	71.47	67.78	68.58
w/o Modality Mixing	71.80	71.66	67.77	68.54
w/o Feature Mixing	71.81	71.66	67.68	68.40
One Mixer Block	71.19	71.04	67.80	68.51
w/o Auxiliary Loss	71.06	70.92	67.47	68.35

Table 5: Component ablation results on IEMOCAP and MELD. AG denotes Adaptive Gating. MCA denotes Modality-centric Cross-modal Alignment. AMM denotes Axis-wise Multimodal Mixer. EPIRC denotes Explicit Pairwise Interaction Residual Correction.

highest overall ACC (68.36%) and WF1 (67.62%), improving over CSS by 1.11% and 1.34%, respectively. Besides, it also achieves the strongest F1 score across all seven emotion categories. The largest gains over CSS occur for fear and disgust, with improvements of 10.29% and 4.96%. These two minority emotions are particularly difficult for existing methods, several of which obtain near-zero or substantially lower F1 scores. The results suggest that preserving axis-specific information while injecting explicit pairwise interactions helps retain weak but discriminative multimodal cues. Nevertheless, the absolute performance on fear and disgust remains below that of the majority emotions, highlighting the remaining challenge of robust minority-class modeling.

Downstream model size and computation. Tables 3 and 4 report the parameter counts and matrix/convolution FLOPs of the evaluated implementations, respectively. We count one multiply-accumulate as two FLOPs, excluding pretrained feature extraction, backward computation, and non-matrix operations such as normalization, activations, and element-wise/scatter message aggregation. MM-Mixer uses 6.36M and 5.71M registered downstream parameters and requires 69.00M and 68.72M FLOPs per utterance on IEMOCAP and MELD, respectively. Compared with CSS, SDT, and Ada2I, it reduces matrix/convolution FLOPs by 65.7%, 76.2%, and 44.8% on IEMOCAP, and by 76.0%, 82.2%, and 21.9% on MELD under the adopted counting convention. These implementation-level counts depend on the input representations and batching and do not establish an inference-speed advantage.

Ablation Study

Component ablation. We evaluate the contribution of each major component on IEMOCAP and MELD by removing it from the full model and retraining it in the same training and evaluation settings. The experimental results are shown in Table 5. We can see that removing any component decreases WF1 on both datasets, confirming the effectiveness of these components and demonstrating that they provide complementary rather than redundant contributions.

Modalities	IEMOCAP		MELD	
	WF1	ACC	WF1	ACC
T	67.98	67.90	66.71	67.43
A	54.10	54.41	48.38	52.53
V	33.66	34.44	32.60	41.92
T+A	70.29	70.24	66.81	67.58
T+V	68.76	68.64	66.37	67.01
A+V	56.55	56.62	48.64	52.68
T+A+V	72.16	72.03	68.01	68.77

Table 6: Modality ablation results on IEMOCAP and MELD. All values are percentages.

On IEMOCAP, removing auxiliary loss causes the largest degradation of 1.10%, followed by MCA with a 1.03% decrease. These results highlight the importance of retaining discriminative unimodal supervision and performing Modality-centric Cross-modal Alignment. Replacing the two-block mixer with one block reduces WF1 by 0.97%. Within the mixer, removing latent-sequence, modality, or hidden-feature mixing decreases WF1 by 0.52%, 0.36%, and 0.35%, respectively, confirming that all three-axis mixing contributes to the final representation. Removing the complete Axis-wise Mixer and Explicit Pairwise Interaction Residual Correction reduces WF1 by 0.72% and 0.52%, respectively, further showing that axis-specific interaction and explicit multiplicative correction contribute complementary information.

On MELD, removing the Axis-wise Multimodal Mixer and Explicit Pairwise Interaction Residual Correction produces the largest degradation of 0.96% and 0.91%, respectively. These results demonstrate the importance of organizing multimodal interaction along separate representational axes while retaining an explicit interaction path for pairwise dependencies. Removing latent-sequence, modality, or hidden-feature mixing individually decreases WF1 by 0.23%, 0.24%, and 0.33%, while using only one mixer block causes a 0.21% decrease. Removing Feature Gating lowers WF1 by 0.26% on IEMOCAP and 0.34% on MELD. Removing adaptive gating also decreases WF1 by 0.48% and 0.65% on the two datasets, indicating that both local feature calibration and context-conditioned refinement are useful. Finally, removing MCA or Unimodal losses degrades performance, confirming the complementary roles of modality-centric alignment and unimodal supervision.

Modality contribution. Table 6 shows the evaluation results on all unimodal, bimodal, and trimodal input combinations. Text provides the strongest unimodal performance on both datasets, whereas audio and visual information alone is substantially less discriminative. Nevertheless, adding audio to text improves WF1 by 2.31% on IEMOCAP and by 0.10% on MELD. The complete T+A+V configuration achieves the strongest overall performance on both datasets, exceeding the best bimodal configuration by 1.87% and 1.20% WF1, respectively. These results show that MM-Mixer can effectively utilize textual, acoustic, and visual evidence for emotion recognition, rather than relying exclusively on the dominant textual modality.

Future Work

The results on MELD indicate that recognizing minority emotions remains challenging despite the overall performance. Future work will examine the error patterns of these classes and investigate how multimodal evidence can better support their recognition.

The current evaluation trains and tests MM-Mixer within IEMOCAP and MELD separately, leaving its cross-dataset generalization untested. We plan to evaluate transfer across datasets under compatible emotion label definitions and consistent feature extraction settings to assess whether the learned fusion structure generalizes beyond the training domain.

Conclusion

We presented MM-Mixer, a framework for multimodal emotion recognition in conversation. Its core Axis-wise Multimodal Mixer expands utterance-level modality representations into learned projection views and separately performs within-modality view mixing, cross-modal exchange, and hidden-feature transformation. Adaptive gating and modality-centric alignment prepare the representations for this interaction, while query-based fusion and a low-rank pairwise residual form the final multimodal representation. Experiments on IEMOCAP and MELD show competitive recognition performance with fewer downstream parameters and lower matrix/convolution FLOPs than CSS, SDT, and Ada2I.

Supplementary Material

A1. Brief Introduction to Baselines

DialogueRNN. This sequence-based ERC model (Majumder et al. 2019) uses recurrent networks to track the global dialogue state, speaker states, and emotion states. It explicitly accounts for speaker identity when propagating conversational context.

MMGCN. This method (Hu et al. 2021) constructs a multimodal fused graph and uses graph convolution to capture multimodal dependencies together with inter-speaker and intra-speaker contextual relations.

MM-DFN. This model (Hu et al. 2022) introduces a graph-based dynamic fusion module that captures contextual dynamics in different semantic spaces. Its fusion process is designed to reduce redundant information and enhance complementarity among modalities.

M3Net. This approach (Chen et al. 2023) combines multivariate and multi-frequency graph propagation to model multimodal conversational relations. The multi-frequency branch captures emotion discrepancy and commonality at different graph frequencies.

Ada2I. This framework (Nguyen et al. 2024) addresses modality imbalance through Adaptive Feature Weighting and Adaptive Modality Weighting. These modules perform feature-level and modality-level balancing by leveraging intra-modal and inter-modal interactions.

SDT. This model (Ma et al. 2024) combines intra-modal and inter-modal Transformer encoders with gated multimodal fu-

sion. It additionally uses unimodal self-distillation to transfer supervision from the fused prediction to the modality-specific branches.

CSS. This framework (Lyu et al. 2026) uses Synergistic Polynomial Fusion with low-rank tensor composition to model higher-order multimodal interactions. Its Pareto Gradient Modulator coordinates competing learning objectives through Pareto-optimal update directions.

A2. More Implementation Details

Input representations and context

We use pre-extracted RoBERTa-large (Liu et al. 2019), wav2vec 2.0 (Baevski et al. 2020), and DenseFace (Zhao et al. 2018) representations for text, audio, and vision, respectively. Their input dimensions are 1024, 1024, and 342. The textual representation of the target utterance includes its preceding dialogue context, whereas the acoustic and visual representations describe the target utterance. We use the full dialogue history, up to 109 preceding utterances on IEMOCAP and 32 on MELD.

Architecture Details

All modality features are projected to a common width of 256. Modality-centric Cross-modal Alignment and learnable-query pooling use eight attention heads. The Axis-wise Multimodal Mixer constructs six learned elements per modality and uses two mixer blocks. Its hidden-feature MLP has a hidden width of 1536. The Explicit Pairwise Interaction Residual Correction uses an interaction rank of 32.

Optimization Details

We optimize MM-Mixer with AdamW. The learning rate is 3×10^{-5} on IEMOCAP and 4.17×10^{-5} on MELD, with weight decay values of 2×10^{-4} and 9.98×10^{-5} , respectively. The batch size is 32, and the maximum number of epochs is 100 on IEMOCAP and 50 on MELD. The fusion dropout is 0.2 on both datasets. We report the mean and standard deviation over three independent runs. The PolyLoss parameters are $\alpha_p = 1.951$ and $\gamma_p = 1.402$, and the focal exponent is $\gamma_f = 2.5$. On IEMOCAP, the weights of the fused, textual, acoustic, and visual objectives are 0.45, 0.33, 0.11, and 0.11. Its main objective uses class weights and a detached batch-level focal coefficient shared by all samples in a mini-batch. On MELD, the fused and three unimodal objectives are equally weighted, and its unweighted main classification loss uses differentiable sample-wise focal coefficients. The auxiliary modality objectives use class-balanced PolyLoss on both datasets.

B1. Visualization Analysis

Component Ablations

Figure 4 and Figure 5 compare Full MM-Mixer with the component-ablation variants using the same test utterances and t-SNE settings. On IEMOCAP, the full model forms relatively compact regions for sadness and anger. On MELD, joy and surprise also form recognizable class-specific regions.

Removing individual components alters these local neighborhoods and weakens their organization, consistent with the performance drops reported by the quantitative ablations in the main paper.

Comparison with Baselines

Figure 6 compares the final fused representations of the reproduced baselines and MM-Mixer on IEMOCAP and MELD. Relative to several recurrent and graph-based baselines, MM-Mixer exhibits more coherent class-wise organization. On IEMOCAP, sadness and anger form compact and distinguishable regions, while on MELD, joy and surprise show identifiable class-specific structures. These qualitative patterns are consistent with the ACC and WF1 comparisons reported in the main paper.

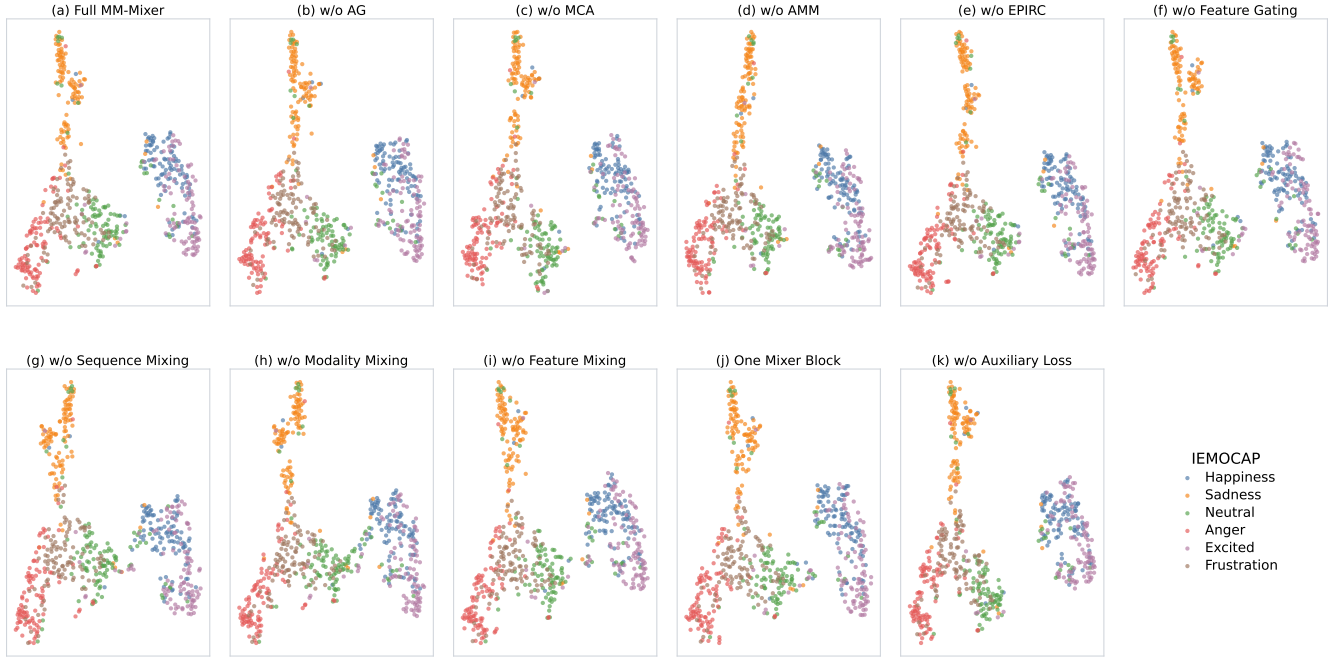


Figure 4: Component-ablation t-SNE visualization on IEMOCAP. “w/o” denotes “without.” AG, MCA, AMM, and EPIRC denote Adaptive Gating, Modality-centric Cross-modal Alignment, Axis-wise Multimodal Mixer, and Explicit Pairwise Interaction Residual Correction, respectively. “Sequence Mixing” in the panels refers to mixing along the learned projection axis, not observed temporal frames. Colors denote ground-truth emotions.



Figure 5: Component-ablation t-SNE visualization on MELD. The variants, sample-selection protocol, preprocessing, and abbreviations follow Figure 4.

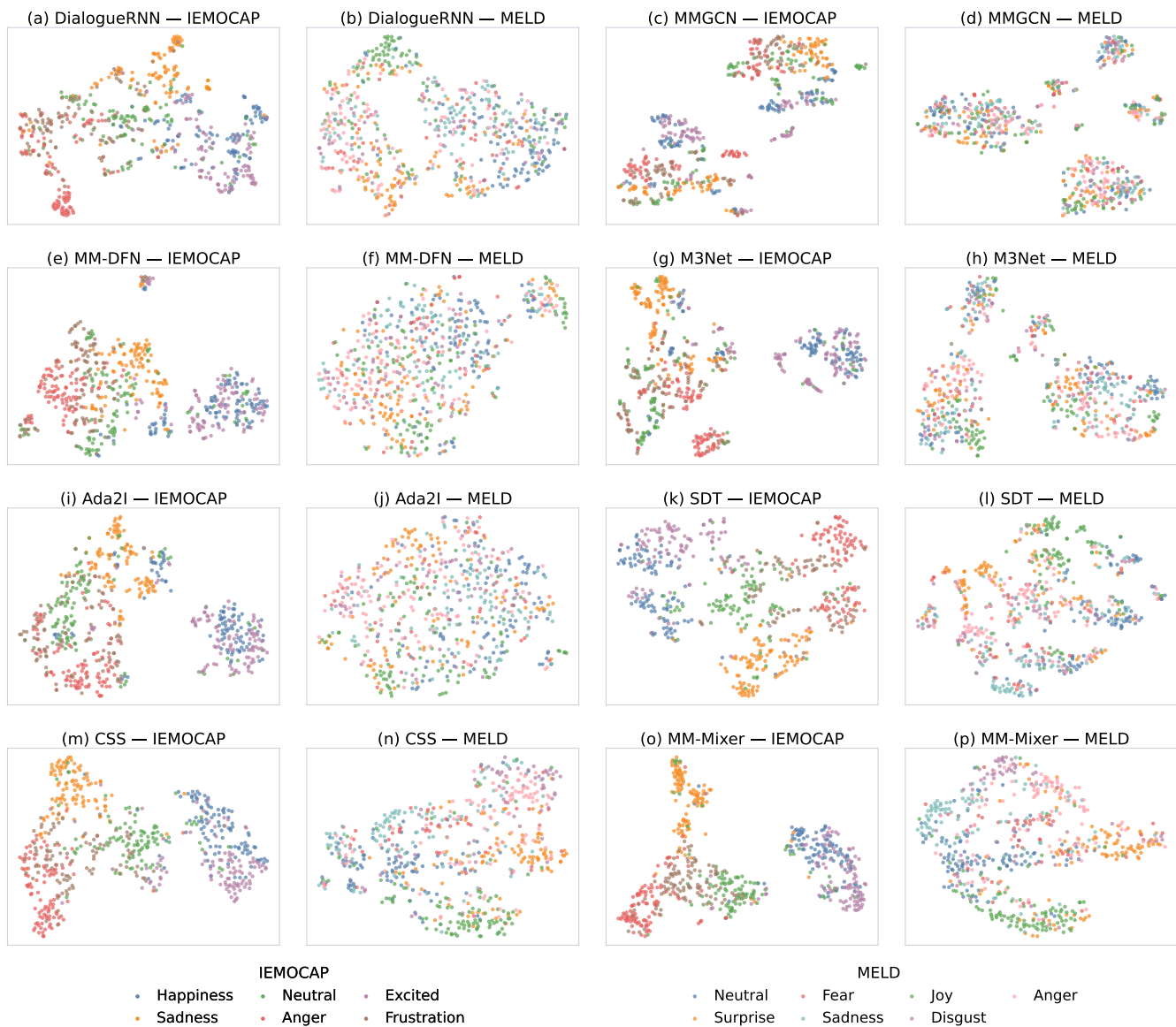


Figure 6: t-SNE visualization of the final fused representations learned by eight models on IEMOCAP and MELD. For each model–dataset pair, class-stratified test samples with the same per-class cap and visualization settings are used. Colors denote ground-truth emotion labels.

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